# Parag Sarvoday Sahu

**Research Internship in 3D Computer Vision & Neural Rendering**

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- Master's student at École Polytechnique working on 3D vision, inverse rendering and Gaussian splatting.
- First-author manuscript in preparation, Pacific Graphics 2025 paper, 2nd place SIGGRAPH Asia 2025 3DGS Challenge.
- Core skills: differentiable rendering and PyTorch research code; available 22 March to 31 August 2027.

## Education

**École Polytechnique (IP Paris), Palaiseau, France** — Sep 2026 – Present
*MSc&T in Visual and Creative Artificial Intelligence* — French Government Scholar (Charpak)
*Courses:* Image Analysis & Computer Vision, Digital Representation & Analysis of Shapes, Computer Animation, Foundations of ML

**Indian Institute of Technology (IIT) Gandhinagar, India** — 2022 – 2026
*B.Tech in Electrical Engineering, Minor in Computer Science* — CPI **9.04**/10, Dean's List
*Courses:* Computer Vision, Machine Learning, Matrix Methods for ML, Probability & Statistics

## Publications

**TensoIS: A Step Towards Feed-Forward Tensorial Inverse Subsurface Scattering for Perlin Distributed Heterogeneous Media**
A. Tiwari, S. Bhardwaj, Y. Bachwana, **P. S. Sahu**, T. M. F. Ali, B. Chintalapati, S. Raman — *Pacific Graphics 2025 (CGF)*

**Manuscript on Multi-View Photometric Stereo with 2D Gaussian Splatting**
**P. S. Sahu**, A. Tiwari, V. Shah, S. Raman — *First author, in preparation for CVPR 2027*

## Research Experience

**IndiaAI Research Fellow (National Merit Fellowship)** — Jan 2026 – Present
*IIT Gandhinagar | Inverse Rendering with Gaussian Primitives* — Advisor: Prof. Shanmuganathan Raman

- Formulated a novel 2D Gaussian Splatting approach to uncalibrated Multi-View Photometric Stereo (MVPS), jointly recovering geometry and materials without known lighting directions or intensities.
- Recovered surface normals at **6.09°** mean angular error on DiLiGenT-MV and supported view-unaligned captures across varying illuminations.
- Cut the required light captures **10**× (10 instead of 96 per view); each object optimizes in under 1 hour on one GPU, versus about 12 hours for PS-NeRF.

**Research Intern, Jodhpur, India** — Jul 2025 – Dec 2025
*3DVisLab, IIT Jodhpur* — Advisor: Prof. Avinash Sharma

- Explored learning-based 3D reflective-symmetry detection as a basis for symmetry-aware mesh compression.
- Applied mesh processing and deep learning to high-fidelity 3D human head reconstruction.

## Competitions & Projects

**SIGGRAPH Asia 2025 3DGS Challenge** — Oct 2025 – Nov 2025
*Real-time 3D Gaussian Splatting for novel view synthesis* — paragsarvoday/DeadlineDinosaur

- Ranked **2nd of about 50 teams worldwide** with **28.04 dB** mean PSNR over 30 test scenes, using a coarse-to-fine progressive-resolution schedule that trains each scene in under 60 seconds.
- Won a **$2,000 USD** prize and presented our method in person at the SIGGRAPH Asia 2025 3DGS Workshop in Hong Kong, covering our backbone choice and hyperparameter study.

**Vision Transformer Inference Engine from Scratch** — Mar 2026 – Apr 2026
*Pure C++ / Eigen, deployed to the browser with WebAssembly* — paragsarvoday/ViT-Inference

- Implemented ViT inference without a deep-learning framework, shipping an on-device web demo.

## Languages

English: fluent (working language) · Hindi: native · French: A1 (CEFR), currently studying

## IT Skills

**Programming:** Python, C/C++, WebAssembly, MATLAB
**Libraries & Tools:** PyTorch, gsplat, Mitsuba 3, Eigen, OpenCV, Git/GitHub, LaTeX

## Honors and Awards

- **Charpak Master's Scholarship:** Selective French government award for graduate studies (1 of 19 nationwide). 2026
- **Bipin & Rekha Shah & Prof. DV Pai Scholarships:** Awarded for academic excellence at IIT. 2023, 2024

## Interests

Badminton · Reading (public bookshelf ↗)